

SRICHARAN KORIDE

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EDUCATION

University of Southern California, Los Angeles, CA

June 2024 – December 2025

Master of Science in Computer Science

CGPA: 3.68 / 4.0

Relevant Coursework: Machine Learning for Data Science, Information Retrieval, Operating Systems, Analysis of Algorithms

CVR College of Engineering, Hyderabad, India

August 2018 – May 2022

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.18 / 10.0

WORK EXPERIENCE

Full Stack Software Engineer (Full-time) | Accolite Digital Private Limited

Hyderabad, India

Client - FedEx, Memphis, TN

May 2022 – May 2024

- Reduced application deployment time from 15 to 5 minutes by overhauling a stalled Jenkins CI/CD pipeline, taking ownership of the project to implement multi-environment support and automate the full application lifecycle
- Orchestrated the end-to-end migration of the authentication layer for two enterprise applications from legacy LDAP to a modern Okta-based OAuth 2.0 framework, fulfilling a critical security and compliance requirement for a larger company-wide transition to Co-location servers
- Built the new JWT verification system using a Spring Security filter, which enabled a centralized, automated audit trail for all user logins via Okta and significantly improved security monitoring capabilities
- Delivered a secure REST API for file uploads allowing pilots to attach documents directly to ticket comments, which replaced a confusing legacy workflow and led to clearer communication and faster issue diagnosis
- Re-engineered two critical legacy modules from AngularJS to modern Angular, successfully replicating complex business workflows and establishing a more maintainable and scalable frontend architecture for future development

Software Development Intern | Accolite Digital Private Limited

Hyderabad, India

Client - FedEx, Memphis, TN

August 2021 – May 2022

- Replaced an unreliable LDAP integration with Workday API for a critical user-status batch job, enabling accurate detection of inactive employees and automated ticket reassignment. Implemented token-based authentication and JSON parsing for all data retrieval
- Tackled a wide range of software defects across the full stack, resolving approximately 20% of all tracked bugs by fixing backend errors, UI defects, and critical warnings identified by SonarQube
- Resolved a critical bug in the platform's printing functionality by engineering a new html2canvas-based PDF export solution, which handled multi-page slicing and ensured reliable, cross-browser document generation without freezing the UI

ACADEMIC PROJECTS

SmartDoc Finder (Spring Boot, FastAPI, Next.js, Docker, RabbitMQ, PostgreSQL, FAISS)

June 2025 – July 2025

- Architected and deployed a full-stack, event-driven RAG search engine using a microservices architecture with Java (Spring Boot), Python (FastAPI), and Next.js; containerized the entire application with Docker Compose and orchestrated asynchronous indexing via RabbitMQ
- Implemented a 3-stage retrieval pipeline featuring hybrid search (Apache Lucene & FAISS), followed by a cross-encoder re-ranker to maximize context relevance before generating answers with a locally-hosted LLM (Ollama), ensuring 100% data privacy
- Validated retrieval effectiveness using the BEIR (scifact) benchmark, achieving a strong nDCG@10 score of 0.654, quantitatively proving the system's ability to accurately find and rank relevant information for complex queries

Weenix - A UNIX like Operating System (C, Operating Systems)

January 2025 – May 2025

- Engineered the core kernel of a UNIX-like OS, implementing robust modules for process/thread management, scheduling, and thread-safe synchronization using mutexes
- Developed a flexible virtual file system (VFS) with support for multiple underlying file systems (RamFS, S5FS), providing a unified and reliable interface for all file operations
- Designed and built a virtual memory manager to handle user address spaces and system calls (fork(), read(), write()), enabling efficient and protected user process execution

Image Classifier - USC (Python, Keras, TensorFlow, Scikit-learn, Matplotlib)

December 2024 – December 2024

- Developed and trained a multi-class image classifier on a dataset of over 11,000 images, leveraging transfer learning with pre-trained CNNs to achieve ~90% accuracy and strong precision, recall, and F1 scores.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, SQL

Frameworks & Libraries: Spring Boot, FastAPI, Angular, Next.js, Keras, TensorFlow

Developer Tools & Platforms: Git, Docker, Jenkins, RabbitMQ, PostgreSQL, FAISS (Facebook AI Similarity Search), WebLogic